STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 6: System of Particles and Rotational Motion

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Define centre of mass. [1]
2. Torque magnitude? [1]
3. SI unit of moment of inertia? [1]
4. Rotational kinetic energy? [1]
5. Condition for angular momentum conservation? [1]
6. Rolling relation? [1]
7. Static rigid-body equilibrium? [1]
8. Parallel-axis theorem? [1]

Section B 2-mark questions - 6 x 2 = 12 marks

9. 2 and 3 kg at x=0 and 5 m: x_cm? [2]
10. 4 N perpendicular at .5 m: torque? [2]
11. I=3 kg m^2, omega=4: rotational energy? [2]
12. Radius .4 m wheel at 8 m/s: omega? [2]
13. Torque 12 N m and I=4: ? [2]
14. No external force: total momentum? [2]

Section C 3-mark questions - 4 x 3 = 12 marks

15. 2 kg at x=0 and 6 kg at x=4: x_cm? [3]
16. .25 m spanner, 40 N at 30: torque? [3]
17. Solid disc M=4 kg, R=.5 m: I and KE at omega=10? [3]
18. omega rises 2 to 10 rad/s in 4 s: and theta? [3]

Section D 4-mark questions - 2 x 4 = 8 marks

19. 2 kg at x=0 and 3 kg at x=5 m, external force 10 N: x_cm and acceleration? [4]
20. Skater I changes 4 to 1.6 kg m^2 from omega=3: final omega and KE change? [4]

End of question paper.

FORMULA AND EQUATION REFERENCE

Chapter-specific equations worth knowing

- Centre of mass on a line: x_cm=(sum m_i x_i)/(sum m_i).
- Torque magnitude: tau=rF sin(theta); rotational equilibrium requires sum tau=0.
- Moment of inertia: I=sum m_i r_i^2; solid disc about centre: I=(1/2)MR^2.
- Rotational kinetic energy K=(1/2)I omega^2; angular momentum L=I omega.
- Angular kinematics: alpha=(omega-u_omega)/t; theta=omega_i t+(1/2)alpha t^2; theta=((omega_i+omega_f)/2)t.
- Rolling without slipping: v=R omega. Parallel-axis theorem: I=I_cm+Md^2.
- Angular momentum conservation: I_i omega_i=I_f omega_f when external torque is zero.
- Translation: a=F_net/M for the whole system.

Use each equation only when its assumptions apply; keep units consistent.

STEP-BY-STEP SOLUTIONS

Solutions are grouped here after the question paper and formula reference.

Question 1

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Mass-weighted mean position.
Final answer: Mass-weighted mean position.

Question 2

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: rF sintheta.
Final answer: rF sintheta.

Question 3

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: kg m^2.
Final answer: kg m^2.

Question 4

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Iomega^2.
Final answer: Iomega^2.

Question 5

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Net external torque is zero.
Final answer: Net external torque is zero.

Question 6

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: v=romega.
Final answer: v=romega.

Question 7

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: F=0 and tau=0.
Final answer: F=0 and tau=0.

Question 8

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: I=I_cm+Md^2.
Final answer: I=I_cm+Md^2.

Question 9

Working: x_cm=(m1x1+m2x2)/(m1+m2)=[2(0)+3(5)]/(2+3)=15/5=3 m.
Final answer: 3 m.

Question 10

Working: Torque magnitude tau=rF sin90=0.5(4)=2 N m.
Final answer: 2 N m.

Question 11

Working: K_rot=0.5 I omega^2=0.5(3)(4^2)=1.5(16)=24 J.
Final answer: 24 J.

Question 12

Working: For rolling without slipping, omega=v/r=8/0.4=20 rad/s.
Final answer: 20 rad/s.

Question 13

Working: Angular acceleration alpha=tau/I=12/4=3 rad/s^2.
Final answer: 3 rad/s^2.

Question 14

Working: With zero net external force, momentum of the complete system remains constant: total p_initial=total p_final.
Final answer: Constant.

Question 15

Working: x_cm=[2(0)+6(4)]/(2+6)=24/8=3 m.

Final answer: 3 m.

Question 16

Working: tau=rF sin(theta)=0.25(40)sin30=10(0.5)=5 N m.

Final answer: 5 N m.

Question 17

Working: For a solid disc I=0.5MR^2=0.5(4)(0.5^2)=0.5 kg m^2. K=0.5I omega^2=0.25(100)=25 J.

Final answer: I=.5 kg m^2; K=25 J.

Question 18

Working: alpha=(10-2)/4=2 rad/s^2. Angular displacement=average angular speed x time=((2+10)/2)(4)=6(4)=24 rad.

Final answer: 2 rad/s^2; 24 rad.

Question 19

Working: x_cm=[2(0)+3(5)]/5=3 m. Total mass=5 kg, so acceleration of the centre of mass a=F/M=10/5=2 m/s^2.

Final answer: 3 m; 2 m/s^2.

Question 20

Working: With no external torque, I_i omega_i=I_f omega_f. Thus omega_f=(4)(3)/1.6=7.5 rad/s. K_i=0.5(4)(3^2)=18 J; K_f=0.5(1.6)(7.5^2)=45 J, an increase of 27 J supplied by the skater's work.

Final answer: omega=7.5 rad/s; K increases 18 to 45 J because skater does work.